

**What Role Does Manufacturing-Led Industrialization Play in
Reducing Regional Development Disparities within Karnataka?**

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Abstract

Manufacturing-led industrialization is usually seen as a very effective and quick way to reduce regional inequality and bring even development. In Karnataka, development is still uneven. Growth and high income opportunities are mainly concentrated in and around Bengaluru and a few key southern districts. Many districts located in north and central regions of Karnataka continue to lag on jobs, industry presence, and incomes. (Shiddalingaswami, H., & Raghavendra, V. K. (n.d.). *Regional disparities in Karnataka: A district level analysis of growth and development* (CMDR Monograph Series No. 60). Centre for Multi-Disciplinary Development Research. <https://cmdr.ac.in/wp-content/uploads/monographs/Mono-60.pdf>)

This paper mainly enquires and analyses how far manufacturing growth can reduce these regional development gaps within Karnataka. The paper is based on secondary data to study and make district-level comparisons. Government statistics and policy documents and case studies of industrial hubs and clusters (for example, Bengaluru region and selected industrial cluster districts) have been used to understand the impact of manufacturing industries in the growth of local employment opportunities and incomes. (Ministry of Statistics and Programme Implementation. (2025, August 27). Annual Survey of Industries (ASI) results for 2023–24. Press Information Bureau, Government of India.

<https://www.pib.gov.in/PressReleasePage.aspx?PRID=2161192>) The data and results suggest that **districts with stronger manufacturing industrial setup and activity have better employment opportunities and more local income growth.** This is mainly because Micro, Small, and Medium Enterprises (MSMEs) create direct jobs for local people, support local suppliers, and increase demand for services like transport, housing etc. According to recent estimates, manufacturing contributes around one fifth of total GDP that makes it a major part of

Karnataka's economy. Due to this location and distribution of these industries matters for balanced growth and reducing disparity .Balanced growth in lagging regions can be achieved by providing proper policy support , developing necessary infrastructure to support manufacturing industries, MSMEs and employable skills to local people of regions that lags in growth.

<https://prsindia.org/budgets/states/karnataka-budget-analysis-2024-25>).Overall Manufacturing-led industrialization can help in reducing regional gaps in Karnataka, but only when growth is decentralized and expanded to beyond existing hubs and linked to local jobs

Keywords: Manufacturing, Industrialization, Regional Disparities, Karnataka, Inclusive Growth, MSMEs

Introduction

Karnataka is one of the economically advanced southern states of India. However a clear disparity or uneven development can be seen in southern karnatka and northern karnatka its development is not even. Districts in the south, especially those around Bengaluru,Mysuru, Ramanagara, Mandya, and Tumakuru show higher income levels, better jobs, and stronger infrastructure. On the other hand , many districts in North Karnataka such as Kalaburagi, Yadgir, Raichur, Bidar, and Koppal show lower per capita income, lower job creation , and very limited industrial activity.

According to the Karnataka Economic Survey, the urban region of Bangalore alone contributes around 35–38% of the state's Gross State Domestic Product (GSDP), while several northern districts contribute less than 3% each. It is mainly due to the concentration of public spending, skilled labour, better roads,Better digital connectivity , Better educational and research institutes and strong links to global markets. On the other hand, in the North Karnataka districts,

the penetration of manufacturing and service sector industries is very weak, and most people are heavily dependent on agriculture and informal forms of employment. This is the reason their per capita income is lower and income levels are also unstable. Historically, North Karnataka did not receive sufficient capital investment from the government, which led to slower infrastructure growth. This situation reflects spatial inequality, where growth is concentrated in specific locations rather than shared across regions. Because of limited employment opportunities outside farming and the informal sector in North Karnataka districts, people mostly migrate toward the developed cities of Southern Karnataka. This uneven development raises important concerns about social inequality and long-term regional stability within the state.

Manufacturing-led industrialisation, commonly known as the growth of the secondary sector, mainly refers to the growth of factories, industrial units, and production activities, and it is considered the engine of economic development. Unlike the services sector, manufacturing industries have a strong ability to create jobs for both skilled and semi-skilled workers. It also supports the growth of other sectors of the economy, such as networks of suppliers, transport services, repair units, and small businesses. This helps in achieving inclusive growth, especially in regions where high-skill service jobs are limited. Manufacturing involves adding value through the process in which raw materials are converted into higher-value and more useful products. The establishment of manufacturing industries in less developed parts of the state helps to increase the local income of people and reduce dependence on agriculture. In this way, the growth of small manufacturing units, small factories, and MSMEs helps to spread the benefits of the local economy instead of concentrating wealth in a single district or region.

Thus, manufacturing-led industrialisation can be seen as a practical tool to reduce regional gaps and promote balanced and inclusive development in the state. This study is an important

topic of discussion because regional inequality is a major challenge in many states, not only in Karnataka. Despite strong overall economic growth, a few regions continue to lag behind. Understanding why the benefits of development and economic growth are concentrated in only a few districts helps in identifying gaps in government policies, planning, and their implementation. This topic studies how unequal industrial and employment opportunities shape income differences across regions. It also highlights the importance of balanced regional development for social equality, reducing migration pressure, and ensuring long-term economic stability. This study examines the role of manufacturing-led industrialization in reducing regional development disparities within Karnataka. The main research question asks whether the growth of manufacturing has helped narrow income and employment gaps between developed and less-developed districts. The study aims to analyse district-level patterns of manufacturing growth, compare economic outcomes across regions, and understand how industrial activity affects local development. This research is important because it helps policymakers assess whether industrial policies are benefiting backward regions or mainly strengthening existing industrial hubs. The findings can guide better planning of industries, infrastructure, and skill development to promote balanced and inclusive growth across the state.

Literature Review

Process of industrialisation and its impact in the regional development has been discussed in many economic theories. Where the balanced growth and unbalanced truth has always been the topic of debate. Balance growth mainly refers to the idea that investment should be made in all the regions and all the sectors in even matter so that the benefit of development is shared among all and majority of the people. In contrast, **Albert Hirschman** argued for **unbalanced**

growth, where investment is first concentrated in selected sectors or regions to create strong growth impulses. According to Hirschman, these leading sectors then create linkages that pull other regions and activities forward over time.

Economist Gunnar Myrdal used the idea of **cumulative causation** to explain regional inequality. This theory helps to understand the continued regional inequality even when the overall economic growth happens. He argued that economic growth often keeps happening and returning to already developed regions mainly due to better infrastructure, skilled workers, and more private and public investments. As a result, some regions continue to grow faster compared to other regions. Poorer regions, on the other hand, lose workers and capital to richer areas. He termed this process as the **backwash effect**. Although some benefits may spread to nearby regions which are known as **spread effects**. In most cases the backwash effects are stronger.

The growth pole theory also supports the idea of cumulative causation. The theory mainly states that development works begin from large industrial or urban centres of a region which becomes centres of innovation and job creation. Economist Paul Krugman also suggested that mostly firms are clustered in specific locations because of advantages of economies of scale, ease of access to factors of production, good market access and lower transport cost. Clustering of the firms and industries make them more efficient and offers many advantages however it also increases the regional gap and disparity if the governmental policies are not available to address this issue. Together, these theories show that manufacturing can reduce regional inequality only when growth spreads beyond major centres.

Global examples also show that in the presence of strong government policy, manufacturing led industrialisation can help to reduce regional inequality and disparity. For

instance, in China, the Government planned and made special economic zones and industrial parks to shift manufacturing plants beyond certain coastal cities. Expansion of manufacturing facilities plants created large scale employment opportunities in poor regions and interior provinces. Similarly in South Korea the government promoted expansion of Industrial and manufacturing complexes outside Seoul to reduce overcrowding, congestion and regional imbalance. Expansion of heavy industries and manufacturing created very strong backward and forward linkages that help local suppliers, logistic companies and other service sector industries to grow. Over time this reduced the income gaps between the regions.

Vietnam, Another South Eastern country that promoted export oriented manufacturing industries in less developed regions through industrial zones. These industrial zones created spill over effects and increased demand for housing, transportation and other local services. Spillover effects not only increase the rural income but also help in reducing the poverty in the surrounding districts however Mini studies suggested that regions without good infrastructure benefited very less. Global literature agrees that manufacturing industries can support regional equality and create job opportunities for low skill and medium skilled people if good Infrastructures support is provided by the government.

In the context of India establishment and growth of manufacturing lead industries is seen as an important tool for inclusive growth. programs such as make in India AIIMS to increase the GDP contribution coming from the manufacturing sector. Manufacturing is also seen as the manufacturing sector is also seen as the sector with stronger job potential For Semi skilled workers which is crucial for the development of backward regions. Studies on industrial corridors, such as the Delhi–Mumbai Industrial Corridor, show mixed results. While these corridors attract large investment and infrastructure growth. But their benefits often remain

concentrated in already developed districts. The benefit of the growth and investment is not evenly spread to the entire region. Several studies also point to a clear north–south divide across Indian states, where southern and western states attract more manufacturing investment than eastern and northern regions.

Studies based on state level industrial policy also show that States with better industrial policy, better Logistic facilities and policies that promote skill based education and employment are performing better in manufacturing growth. However, even within the states such as Karnataka which are registering good economic growth district level desperation still a challenge this suggests that manufacturing growth of manufacturing LED Industries alone is not enough. The location, scale, and local linkages of industries matter greatly. Indian literature therefore stresses the need for decentralized industrial strategies and MSME support to ensure balanced regional development.

Available research and data provides strong theoretical state level and national-level insights. However there is a clear gap in district-level analysis within Karnataka. Most studies focus on state averages or data based on major industrial hubs, especially Bengaluru. Research by CMDR and findings of the Nanjundappa Committee indicate that Bengaluru Urban alone contributes around 35–38% of Karnataka's GSDP, while several North Karnataka districts contribute less than 3% each. Per capita income in southern districts such as Bengaluru Urban and Mysuru is more than two to three times higher than districts like Yadgir, Raichur, and Kalaburagi. The literature consistently links this inequality to the concentration of industry, infrastructure, and formal employment in a few districts, confirming that Karnataka's economic growth has been spatially uneven. There is limited work linking manufacturing growth directly with regional inequality indicators such as district income, employment structure, and human

development. Few studies compare industrialized and less-industrialized districts within the same state. This paper addresses this gap by using district-level data and case studies to examine whether manufacturing-led industrialization has actually reduced regional development disparities within Karnataka.

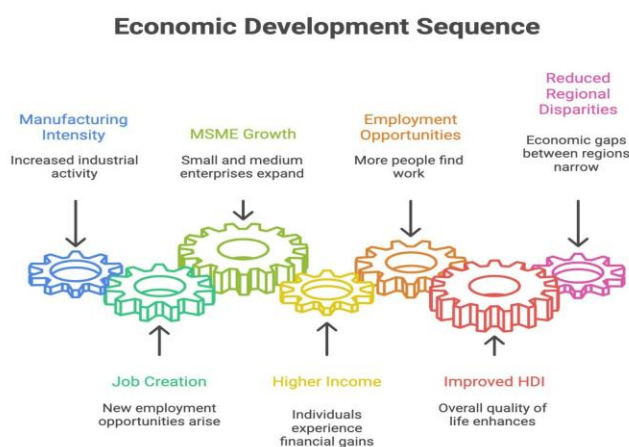
Conceptual Framework

The conceptual framework of this study is based on the idea that **manufacturing-led industrialization can reduce regional development disparities through multiple connected pathways**. The key to reducing regional inequality is to generate employment for skilled, semi-skilled, and unskilled workers. This is especially important in districts where agriculture is the main source of income and job opportunities are limited. Stable jobs in manufacturing industries can help increase household income and reduce seasonal unemployment.

Simultaneously, the growth of micro, small, and medium enterprises (MSMEs) can play an important role in addressing regional disparities. Large manufacturing units often depend on local suppliers for components, packaging, transport, repair, and maintenance services, which creates strong linkages with MSMEs and helps spread income within the local economy. According to the Ministry of MSME (Government of India), the MSME sector contributes around 25-30% of India's GDP and nearly 35-40% of total manufacturing output, while providing employment to over 100 million people. These enterprises are more geographically dispersed than large industries, making them especially important for generating jobs and income in backward and semi-urban regions. As a result, MSME-led industrial growth can support more balanced regional development by reducing over-concentration of economic activity in a few districts. manufacturing creates **urban spillover effects** into nearby rural areas. Towns grow

around industrial clusters, and demand rises for housing, retail, education, and health services. These spillovers help integrate rural areas into regional growth processes and reduce spatial inequality over time. (Ministry of Micro, Small and Medium Enterprises. (2023). Annual report 2022–23. Government of India. <https://msme.gov.in>) Furthermore, investment and development of **infrastructure can help increase in the manufacturing sector**. Manufacturing requires good roads, continuous power, water, and logistics facilities. When industries are set up in less-developed districts, public and private investment in infrastructure usually follows. This improves connectivity and access to markets for the wider population.

This study uses a simple analytical framework to examine the relationship between manufacturing and regional development in Karnataka. The **independent variable** is **manufacturing intensity**, measured through indicators such as number of manufacturing units, industrial employment, and manufacturing output at the district level.



The dependent variables that we are measuring are the main development outcomes, including per capita income, employment levels, and human development indicators (HDI). These indicators capture both economic and social aspects of regional development. To ensure a

balanced analysis, the methods also includes control variables such as education levels, infrastructure availability, and urbanization rates. These factors influence development outcomes of any district alongside manufacturing. By comparing districts with different levels of manufacturing intensity, the framework helps assess whether manufacturing-led industrialization contributes to reducing regional disparities

Methodology

This study uses descriptive and analytical research design. The descriptive part mainly explains the pattern of manufacturing growth and growth in regional development across different districts of Karnataka. It help to show the clear link between industrial activity income levels and reduction in unemployment in various regions of the state. On the other hand analytical part is used to exam in the relationship between manufacturing lead industrialisation and outcomes in the regional development.

The study is based entirely on secondary data, as reliable and long-term data are available through government and institutional sources. A comparative district-level analysis is used to compare more industrialized districts with less industrialized ones. Published data between the time period of 2015 to 2023 has been used to analyse the long term Trend and understand the impact of recent policies This approach helps in understanding whether districts with higher manufacturing intensity perform better in terms of income, employment, and human development. By combining description with comparison, the research design allows a balanced and evidence-based assessment of regional disparities within the state. The study uses data from multiple reliable government and institutional sources to ensure accuracy and consistency. The main source is the *Economic Survey of Karnataka*, which provides district-level information on

income, sectoral contribution, employment, and infrastructure. Manufacturing-specific data are taken from the *Annual Survey of Industries (ASI)*, which gives details on factory numbers, output, and industrial employment.

Employment and workforce data are taken from the *National Sample Survey (NSS)* and the *Periodic Labour Force Survey (PLFS)*, which help in understanding changes in employment structure over time. To capture local industrial activity, data from *District Industrial Centres (DICs)* are also used, especially for information on MSMEs and small manufacturing units. Using multiple sources helps cross-check findings and strengthens the reliability of the analysis.

This research also has some limitations such as. Obtaining the district level data for analysis and comparison was difficult as it is not available for all years. Secondly there are many informal manufacturing activities happening in the state which are not accounted in the records specially which are located in smaller towns and rural areas. Which means the employment level could be more than the reported or recorded employment. Finally, the study mainly identifies **correlation rather than direct causality**. While manufacturing growth is linked with better development outcomes, other factors such as education, infrastructure, and urbanization also play an important role.

Manufacturing Landscape in Karnataka

According to a recent state economics survey and Niti Aayog report, the manufacturing sector in Karnataka contributes around 18 to 20% of Karnataka's Gross State Domestic Product (GSDP). Manufacturing is Second most important sector after the service sector. It generates large share of formal jobs and supports many indirect and informal jobs such as transport trade and services. However its benefits are not evenly shared across the reasons. The sector has been source

of large Employment generation specially in factories MSME'S and other allied activities (NITI Aayog. (2025, March). *Macro and fiscal landscape of the State of Karnataka*. Government of India. <https://www.niti.gov.in/sites/default/files/2025-03/Macro-and-Fiscal-Landscape-of-the-State-of-Karnataka.pdf>)

Karnataka has a **diverse manufacturing base**. In electronics and electronic equipment manufacturing the state is a national leader as it is supported by a very strong supply chain and skilled labour force. Furthermore the state has a strong manufacturing facility for automobile and auto component which mainly fulfills domestic and export demands **Textiles and garments** provide employment to a large semi-skilled workforce, especially women. **Food processing** industries are important in districts with strong agricultural production, as they add value to local raw materials and support rural incomes. Despite this diversity, manufacturing growth has been uneven. High-value and technology-intensive manufacturing is mostly concentrated in and around major cities. Traditional and resource-based industries dominate many interior districts. This uneven structure directly influences income and employment differences across regions.

Geographical distribution of manufacturing industries in Karnataka shows clear concentration around certain reasons for instance Bengaluru urban is the most industrialised district in the state. This region and certain other clusters has the highest number of registered factories MSME with a large number of manufacturing output that generates a lot of employment. The reason for the concentration of manufacturing industries around these districts is availability of skilled labour forces, strong infrastructure, Global connectivity and better access to finance. As a result, manufacturing here is capital-intensive and high value, especially in electronics, precision engineering, and advanced manufacturing. **Mysuru** has emerged as a secondary manufacturing hub. It has industries related to food processing, textiles, and light

engineering. Industrial growth in Mysuru has helped create local jobs and reduce migration pressure. However, its scale remains much smaller than Bengaluru. **Belagavi** is an important manufacturing district in North Karnataka. It has a strong base of MSMEs, especially in auto components, foundries, and machine tools. Manufacturing here provides employment to local workers and supports surrounding rural areas. Even so, limited infrastructure and distance from major markets reduce its growth potential compared to southern districts. In contrast, **Kalaburagi** and **Ballari** show weaker manufacturing presence. Kalaburagi's economy is still largely agriculture-based, with few industrial units. Ballari has some industrial activity linked to minerals and steel, but employment generation is limited and highly cyclical and seasonal in nature. These districts show fewer MSMEs, and weaker industrial growth. As a result, income levels and job opportunities remain low. District-wise comparisons clearly show that manufacturing concentration closely matches regional income and employment patterns. Districts with stronger industrial bases perform better on development indicators, while low-industrial districts continue to lag.

To address the issue of regional imbalance Karnataka government has taken several steps and introduced Karnataka industrial policy to attract investment in manufacturing and its setting manufacturing facilities in less developed areas. The Karnataka Industrial Policy focuses on attracting investment, promoting MSMEs, and encouraging industries to locate in backward districts through incentives. Special support is provided for land availability, power supply, and skill development. This district wise comparison data clearly shows the large regional disparity within Karnataka where Bengaluru urban rank first in composite economic Complex with score of 859,154 far ahead of all districts. In terms of GDP per capita Bengaluru urban is ranked top with 7.38 lakh while districts like belagavi and Bagalkote show much lower per capita income

showing Economic output does not evenly translate to individual prosperity across the districts. This indicates very strong concentration of economic activity, in contrast Northern Karnataka districts such as Kalaburagi (Rank 14), Raichur (Rank 17), and Yadgir (Rank 30) are much lower in composite scores showing the weekend economic activity and weaker growth.

The state is also part of two major Industrial Corridor project to improve the connectivity between the centre ports and market areas. They are Bengaluru–Mumbai Industrial Corridor (BMIC) and the Chennai–Bengaluru Industrial Corridor (CBIC) Led by National Industrial Corridor Development Corporation in coordination with Karnataka state government. The main aim of these corridors is to reduce the Logistic costs and improve the overall infrastructure to attract more investment in manufacturing specially in new backward regions such as Belagavi, Dharwad, Ballari, Tumakuru, and Chitradurga. Irritation to that Karnataka state government has developed Mangalore Special Economic Zone which mainly focuses on manufacturing of pharmaceuticals engineering and Port based industrial items in areas such as Peenya, Vemagal, Harohalli, and Kadechur. Karnataka industrial areas developmental board is also promoting manufacturing outside the capital region within aim to create employment and strengthen the local supply chain to support growth of MSMEs. However evidence suggest that most of the investment continue to concentrate in the areas which are already developed because they have better infrastructure skill labour availability and the stronger industrial ecosystem that helps to provide better returns and better output in investment. While the backward regions face the challenges Of shortage of skilled labour supply poor Logistic networks and very limited access to finance this underlines the need for a better and stronger policy and better implementation that could target the skill development and promote the region specific industrial planning for balanced growth and development within Karnataka

Table: District-wise Composite Economic Ranking

District	Composite Score	Overall Rank
Bengaluru Urban	859,154	1
Dakshina Kannada	125,140	2
Belagavi	91,449	3
Tumakuru	82,626	4
Mysuru	80,676	5
Kalaburagi	45,868	14
Raichur	40,226	17
Yadgir	22,754	30

Case Studies of Manufacturing-Led Regional Development

Bengaluru Urban is the most developed manufacturing district in Karnataka .it shows how manufacturing-led industrialization can support regional growth. In the last 20 years Bengaluru has emerged as a major hub for electronics and electrical manufacturing, including hardware components, precision instruments, and advanced engineering products. Annual Survey of Industries (ASI) and the Karnataka Economic Survey data Show that the number of registered manufacturing units are disproportionately high in Bengaluru district. Bengaluru urban alone contributes 35 to 40% of Karnataka state gross domestic product. As a result the district has emerged as one of the leading hub of Electronics and electrical manufacturing. The expansion of electronics manufacturing has led to strong employment growth. Factories and

allied units provide direct jobs to engineers, technicians, and shop-floor workers. At the same time, indirect jobs have grown in logistics, maintenance, food services, and housing. This has helped raise average household incomes and reduce unemployment within the district. This has led to more consumer spending, which further supports local businesses.

Another important impact is skill development spillovers. electronics and electrical manufacturing, including precision instruments, aerospace components, and advanced engineering products, supported by clusters such as Electronic City, Peenya, and Whitefield. Manufacturing firms invest in training workers in technical skills, quality control, and digital tools. Many workers later move to other firms or start small units of their own. This spreads knowledge within the local economy. Strong presence of Technical Institute Universities and Research Institute also provides better human resource and skilled workforce easily to these manufacturing firms. As a result, the district shows higher productivity, better wages, and stronger human development outcomes. However, Bengaluru's success also highlights a limitation. Manufacturing growth here is highly concentrated and capital-intensive. We can see the concentration of investment has reduced the inequality within the urban areas and the developed districts but it is also leading to the Migration of laborers and pulling away the investments away from the less developed regions which aggravates the regional gaps within the state. The cap may continue to increase if the growth is not decentralized with proper policy support.

Table: Sectoral Dependence by Region

Region	Dominant Sector	Employment Nature
Southern Karnataka	Services & Manufacturing	Formal, skilled
Coastal Karnataka	Services, Tourism	Semi-skilled
North Karnataka	Agriculture	Informal, low- income

Belagavi and the Hubballi–Dharwad region provide an important example of manufacturing-led development outside the main metropolitan core. This region has a strong base of MSMEs, especially in auto components, foundries, machine tools, and light engineering. Unlike Bengaluru, manufacturing here is more labour-intensive and locally rooted. The presence of MSME is has helped to create employment opportunities for Semi skilled and unskilled local workers. It has not only reduce the overcrowding of capital regions and highly developed regions but it has also reduced the migration as the workers come from nearby rural areas and support rural households. Small firms also rely on local suppliers and service providers, which spreads income across the region. Government is also opening Technical Institute and industrial training centres for skill formation in smaller regions. This is helping to develop industrial states and basic infrastructure that can eventually attract the manufacturing industries while income levels in Belagavi and Hubballi–Dharwad remain lower than in Bengaluru, the region performs better

than many other districts in North Karnataka. This shows that manufacturing clusters, even without high-end technology, can reduce regional inequality when supported by MSMEs and employment-focused growth. However, growth is still constrained by logistics, access to finance, and limited large-scale investment.

Kalaburagi and Ballari represent districts where manufacturing-led development has been limited. It is heavily dependent on agriculture and government sector employment. There are very few manufacturing units and most of them are very small in scale and size. as the result the overall employment opportunities are very low and income also remains low .

Ballari has some industrial activity linked to mining and steel, but it is **capital-intensive and cyclical in nature**. It creates fewer jobs relative to output and does not support a wide MSME base. Environmental concerns and regulatory uncertainty have also affected industrial stability. Limited supply of skilled workforce poor transport and communication infrastructure reduces the Industrial Investment In both districts. These conditions contribute to **persistent poverty and outward migration**. Many workers move to Bengaluru and other cities in search of jobs. This further weakens local economies and reinforces regional disparities. The experience of Kalaburagi and Ballari shows that without targeted industrial policy, manufacturing does not automatically spread to backward regions. Infrastructure, skills, and MSME support are essential for manufacturing to play a meaningful role in reducing regional inequality.

Impact Assessment: Has Manufacturing Reduced Disparities?

By looking at income and employment indicators we can SS whether manufacturing lead industrialisation has reduced the regional disparity within the state. Government data show that districts with higher manufacturing activity generally record higher per capita income. Bengaluru

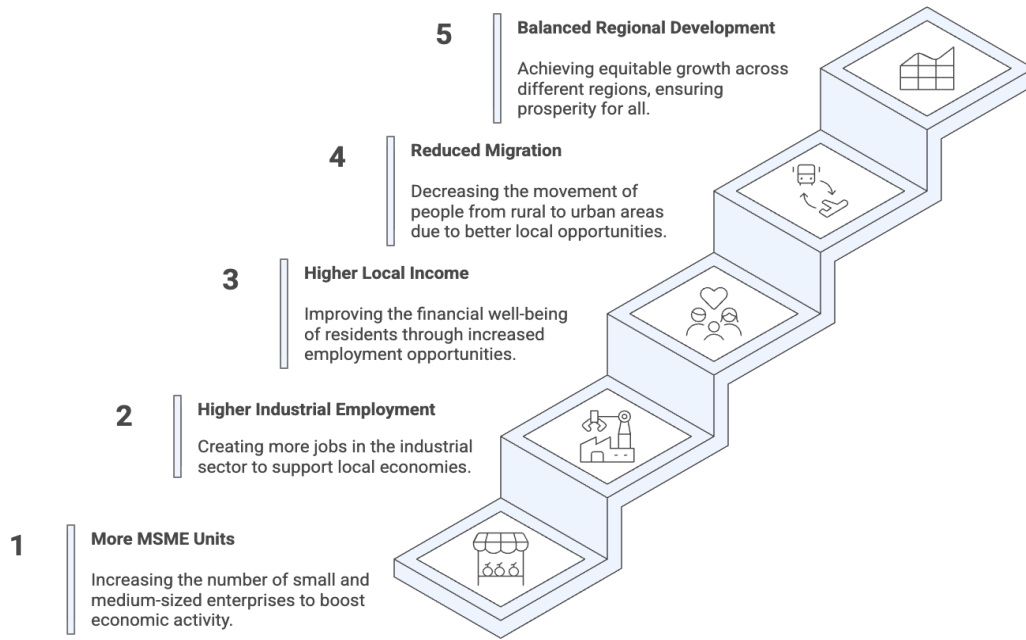
Urban,consistently reports per capita income far above the state average Main did you to better infrastructure and better supply of skilled labours. In contrast, districts such as Kalaburagi record much lower income levels. This pattern suggests a clear link between industrial presence and income generation.

Employment indicators show a similar trend. Districts with more factories and MSMEs have higher workforce participation rates, especially in non-agricultural employment. Manufacturing creates regular wage jobs, which are more stable than seasonal agricultural work. Data from labour surveys show that districts with stronger manufacturing activity have a larger share of workers employed in the secondary sector. This helps reduce underemployment and income insecurity.

The manufacturing employment share is especially important. Manufacturing has higher employment elasticity compared to capital-intensive services. Even moderate industrial growth can create a large number of jobs for semi-skilled and unskilled workers. In districts like Belagavi and Mysuru, manufacturing has helped absorb local labour and reduce migration pressure. However, in districts with limited industrialization, workers continue to depend on low-productivity agriculture or migrate to cities. Overall, income and employment data suggest that manufacturing has reduced disparities within industrialized regions, but its uneven spread limits its impact at the state level. (Department of Economics and Statistics, Government of Karnataka. (2024). Karnataka district-wise MSME industries 2023: District-wise MSME units and employment generated (as of November 2023). OpenCity.

<https://data.opencity.in/dataset/de00ffc3-ac9c-4bab-afb9-e2c09cbf8ba7/resource/54348b35-ca19-414e-9654-89136c474238>)

Achieving Balanced Regional Development



Developed vs Lagging districts.

Table: Top 10 Districts by GDP & Per Capita Income

Rank	District	GDP (₹ lakh crore)	Per Capita Income (₹ lakh)
1	Bengaluru Urban	9.98	7.38
2	Dakshina Kannada	1.37	5.56
3	Belagavi	9.90	1.62
10	Bagalkote	5.80	2.49

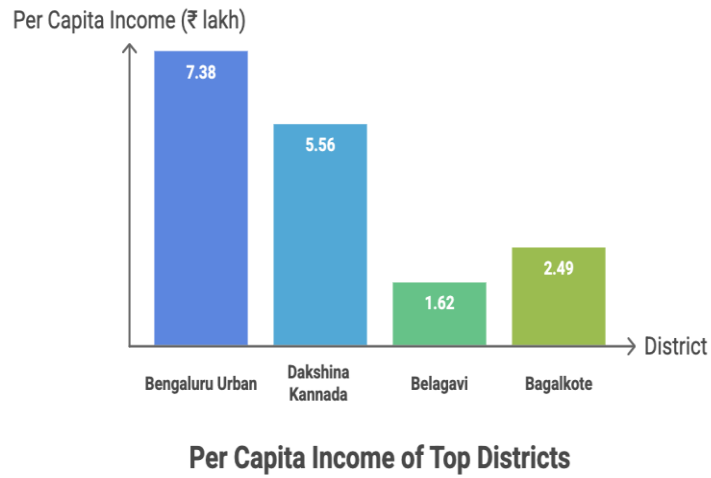
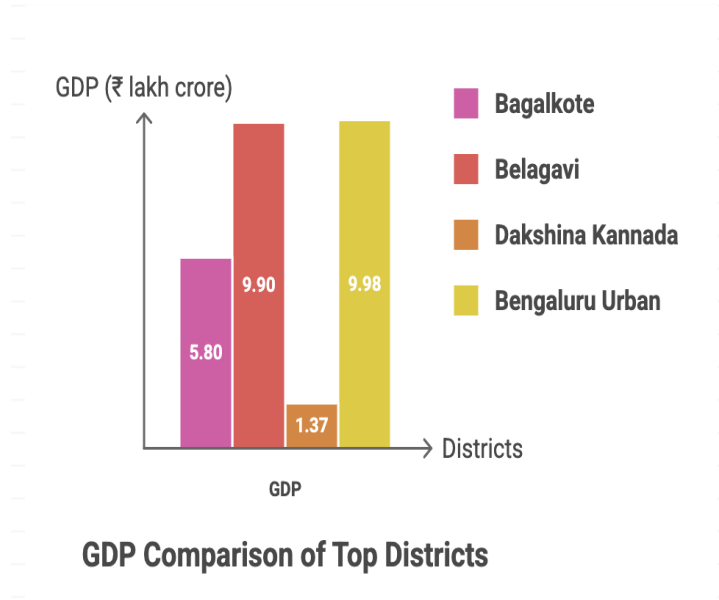


Table: MSME Concentration and Employment Generation (2023)

District	Total MSME Units	Total MSME Employment	Industrial Profile
Bengaluru Urban	323,483	3,071,244	Highly

			industrialised, diversified
Belagavi	82,376	566,756	Strong manufacturing & MSMEs
Dharwad	48,760	368,017	Balanced industrial growth
Kalaburagi	25,424	231,145	Limited industrial base
Koppal	14,746	147,624	Agriculture- dominant
Gadag	15,111	128,900	Low industrial penetration

Human Development Outcomes

In addition to creating better employment and income opportunities, manufacturing also affects human development outcomes such as education, health, and urbanization of areas around where industries are located .Data shows, the Districts which have better infrastructure and industrial activity attract more investment for private Training Institute technical colleges. As a result overall they perform better in terms of education standards and indicators. It also allowed the families with stable household incomes to invest more in better education and skill development. Furthermore, health outcomes also show similar but important links.

Manufacturing growth often brings better infrastructure such as roads, electricity, and healthcare facilities. Urban and semi-urban areas around industrial clusters usually have better access to medical and healthcare services. This contributes to improved health indicators over time. However, benefits are not automatic. Environmental pressure pollution and congestion can offset gains if growth is poorly managed. Moreover data shows growth rate of towns and peripheral urban areas is more around the industrial districts than the districts without manufacturing industries. Quick Urbanizations this creates better access to services and markets for surrounding rural populations. Human development data therefore support the view that manufacturing can improve social outcomes, but only where it is combined with supportive infrastructure and public services.

Comparison between north and South Karnataka supports the view that the growth of industries is uneven and as a result there is clear regional inequality. Southern district closer to Bengaluru have higher income and better employment opportunities and stronger Human Development indicators on the other hand Northern district is far behind in most of the measures. it is directly reflected in the lower per capita income and high poverty levels as in north Karnataka agriculture is still a dominant employer and manufacturing has not grown in a way to change the economic structure

The comparison suggests that manufacturing has reduced disparities within regions where it is present, but it could not fully bridge the gap of uneven development. The main reason is spatial concentration. Manufacturing growth has provided more advantages to already developed regions rather than correcting them.

Challenges and Constraints

Manufacturing-led industrialization can clearly help to reduce regional disparities in Karnataka, but its impact is limited mainly due to several structural and policy-related challenges. One major challenge is infrastructure gaps between advanced districts and backward districts. Many districts in North and central Karnataka lack reliable and continuous power supply, quality roads, rail connectivity(for better logistics), and industrial-grade water availability. Manufacturing firms depend heavily on these basic inputs for faster growth. This leads to overall increase in production costs and reduces competitiveness. As a result, investors prefer districts that already have developed infrastructure, especially around Bengaluru. This creates a cycle where developed regions continue to attract more industry and investment, while backward districts fall further behind.

Another important constraint is skill mismatch. Manufacturing today requires technical skills, machine handling ability, and basic digital literacy. However, skill training facilities and industrial training institutes are unevenly distributed across the state. It is difficult for industries to find suitable and skilled laborers from the local area as backward districts often lack quality vocational training institutes and higher education level institutes. As a result manufacturing industries are forced to import skilled workers from other regions which reduces local employment benefits and weakens the role of manufacturing in reducing regional inequality.

Capital concentration in Bengaluru is another serious issue. Most large investments, financial institutions, and venture capital flows are focused on Bengaluru and nearby districts. This concentration supports high-end and capital-intensive manufacturing but leaves little capital

for smaller towns. MSMEs in backward regions face difficulties in accessing credit, technology, and markets. Without finance and scale, manufacturing in these areas remains small and fragile.

Quick growth in Manufacturing also creates large **environmental costs and pollution**, especially in poorly regulated regions. Industrial pollution, water stress, and land degradation can harm agriculture and health. In mining and heavy industry districts such as Ballari, environmental damage has reduced long-term development benefits. These costs can offset income gains if environmental safeguards are weak. Finally, **policy implementation issues** limit the effectiveness of industrial strategies. While Karnataka has strong industrial policies on paper, execution at the district level is uneven. Delays in land acquisition, slow decision making, and poor coordination between departments reduce investor confidence in backward regions. Strong local governance is needed along with incentives to invest in backward regions. These challenges show that manufacturing can reduce disparities only when supported by infrastructure, skills, finance, and effective policy delivery.

Policy Recommendations

To reduce regional development disparities in Karnataka, manufacturing growth must be actively directed toward backward districts, especially in North and parts of central Karnataka. One key step is the creation of decentralized industrial parks in smaller towns rather than concentrating them near Bengaluru. These parks should offer ready land, power, water, and basic logistics so that firms can start operations quickly. Locating parks closer to raw material sources and labour markets can reduce costs and encourage local hiring.

Improving MSME access to credit is equally important. Small firms in backward regions often struggle to obtain loans due to weak collateral and limited banking presence. Targeted

credit schemes, interest subsidies, and stronger district-level banking support can help MSMEs expand production and employment. Digital lending platforms linked with state guarantees can further reduce risk for lenders.

The state should also invest in skill hubs near industrial parks. These hubs should focus on practical skills such as machine operation, maintenance, quality control, and basic digital tools. Training programs must be linked to local industry needs so that firms can hire locally. This approach can raise employment, reduce migration, and strengthen local manufacturing ecosystems.

A balanced industrial strategy should focus on **cluster-based development**. Instead of spreading resources thinly, the state can identify a few priority industries for each region based on local strengths, such as agro-processing, textiles, or light engineering. Supporting supplier networks within clusters helps firms grow together and improves resilience.

Investment in **logistics and connectivity** is also critical. Better roads, rail links, and storage facilities can reduce transport costs for firms in remote districts. This makes these regions more attractive for investment and helps local products reach wider markets.

Public-private partnerships (PPPs) can support this strategy. Private firms can bring capital and technology, while the state provides infrastructure and coordination. PPPs can also help manage industrial parks and training centers more efficiently. Manufacturing expansion must also be **environmentally sustainable and socially inclusive**. Promoting **green manufacturing** practices such as energy efficiency, waste reduction, and cleaner technologies can reduce environmental damage. Incentives for renewable energy use and water recycling can support long-term growth. At the same time, policies should prioritize **local employment**. Firms

receiving incentives should be encouraged to hire and train local workers. This ensures that manufacturing growth directly benefits local communities and contributes to reducing regional inequality in a lasting way.

Conclusion

The case studies of manufacturing-led regional development in Karnataka clearly show that manufacturing has the potential to promote economic growth, employment generation, and human development, but its impact depends largely on where and how it is distributed. Districts such as Bengaluru Urban demonstrate how strong industrial concentration, supported by infrastructure, skilled labour, and institutions, can raise income levels, reduce unemployment, and improve overall living standards. Manufacturing growth in these areas has created both direct and indirect employment, encouraged skill development, and supported better education and health outcomes.

At the same time, the experience of regions like Belagavi and Hubballi–Dharwad shows that manufacturing does not need to be high-tech or capital-intensive to reduce regional disparities. Labour-intensive MSME-based manufacturing can create employment for semi-skilled and unskilled workers, reduce migration, and spread income within local economies. These regions perform better than other lagging districts in North Karnataka, highlighting the importance of decentralized and employment-focused industrial growth.

In contrast, districts such as Kalaburagi and Ballari illustrate the limitations of uneven industrial development. Dependence on agriculture and capital-intensive activities like mining

has resulted in low employment generation, low incomes, and continued outward migration. The lack of infrastructure, skills, and MSME support has prevented manufacturing from playing a meaningful role in reducing poverty and inequality in these areas.

Overall, manufacturing has reduced disparities within industrialized regions, but it has not fully addressed regional inequality at the state level due to its spatial concentration in already developed districts. For manufacturing to become an effective tool for balanced regional development, targeted industrial policies, improved infrastructure, skill development, and stronger MSME support in lagging regions are essential. Without such measures, manufacturing growth may continue to reinforce existing regional imbalances rather than correct them.

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